

MAS – Final project documentation

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1 User requirements

1.1 Short overview

The system is intended to support the daily operations of a veterinary clinic, covering the full lifecycle of a pet's visit – from scheduling an appointment, through recording performed procedures and prescriptions, to processing the payment. It is used by three types of staff: receptionists, who manage clients, pets, and appointments; veterinarians, who record medical data; and administrators, who maintain the clinic's directories and employee records.

1.2 Functional requirements

1.2.1 People and roles

1. The system must store information about individual clients: first name, last name, date of birth, phone number, and email address.
2. The system must store information about shelters: name, list of phone numbers (at least one phone number must be present), and email address.
3. The system must store information about employees: veterinarians, nurses, and possibly other types in the future. Each employee is characterized by first name, last name, date of birth, phone number, hire date, and hourly rate.
4. Each person in the system must be at least 18 years old.
5. Phone numbers must be unique among individual clients.
6. Phone numbers must be unique among shelters.
7. The same person can be both a client and an employee at the same time.
8. The receptionist must be able to register new individual clients and shelters.
9. The receptionist must be able to edit individual clients' and shelters' information.
10. The administrator must be able to manage (add, edit, remove) employees' data.
11. Only administrator must be able to remove clients and shelters from the system.

1.2.2 Veterinarians

1. Each veterinarian is additionally characterized by a license number, license issuing authority, license issue date, and license expiry date.
2. A veterinarian's license number must be unique.
3. Each veterinarian must have at least one specialization.
4. Each specialization is characterized by a unique name.
5. The directory of specializations must be maintained by the administrator.
6. All authorized staff members must be able to view the list of veterinarians and the schedule of each veterinarian.

1.2.3 Nurses

1. Each nurse must have at least one certification.
2. Each certification is characterized by a unique name.
3. The directory of certifications must be maintained by the administrator.

1.2.4 Pets

1. Each pet is characterized by a name, gender (male, female, or unknown), date of birth, and an optional chip number.
2. Each pet is assigned a unique clinic registration number, used to identify and look up pets within the system.
3. If a pet has a chip number, it must be unique within the system.
4. Each pet belongs to exactly one species.
5. Each species is characterized by a unique name.
6. Each pet must be owned by either an individual client or a shelter.
7. A client or shelter can have zero or more pets.
8. The receptionist must be able to register new pets.
9. The receptionist must be able to edit or delete existing pets' data.
10. The species directory must be managed by the administrator.
11. Each pet must have its medical profile including history of appointments, performed procedures, and prescriptions. Only the veterinarian must be able to access it.

1.2.5 Appointments

1. Each appointment is characterized by a start date and time, end date and time, expected duration, and status.
2. An appointment can have one of the following statuses: scheduled, cancelled, in process, no-show, completed.
3. An appointment can be scheduled with an expected duration of 15 minutes, 30 minutes, 1 hour, 2 hours, or 4 hours.
4. Each appointment is conducted by one veterinarian.
5. Each appointment is for exactly one pet.
6. Appointments of the same veterinarian must not overlap.
7. An appointment's end date and time is recorded only when it is marked as completed.
8. The system must allow receptionists scheduling appointments for pets with veterinarians.
9. Appointments can only be scheduled for time slots within working hours (currently 10:00 to 18:00, subject to change).
10. Appointments can be scheduled up to 90 days in advance (subject to change).
11. After an appointment is scheduled, the system must send a confirmation email to the pet owner's email address.
12. The "in process" status must be assigned automatically once the start time of the appointment has been reached, provided the appointment has not been cancelled.
13. An appointment can be scheduled, and cancelled (only before the appointment start time) by the receptionist; marked as no-show (appointment start time has been reached, but the pet did not appear), and completed by the veterinarian.
14. During marking an appointment as completed, the veterinarian must provide a list of performed procedures (at least one), and optionally a list of prescriptions.

1.2.6 Procedures

1. Each procedure is characterized by a unique name, optional description, and price.
2. Each recorded performed procedure can have notes attached.
3. A procedure can require zero or more specific veterinarian specializations.
4. A procedure can require zero or more specific nurse certifications.
5. The same procedure can be performed across multiple appointments, and multiple times in a single appointment.
6. A veterinarian can only perform a procedure if they hold all specializations required by that procedure.
7. A nurse can assist at a procedure if they hold all certifications required by that procedure.
8. Zero or more nurses can assist at a performed procedure.
9. A completed appointment must have at least one performed procedure.
10. The catalogue of procedures must be maintained by the administrator.
11. During an appointment, multiple procedures from the catalogue can be performed.
12. Each performed procedure must be recorded in the system.

1.2.7 Prescriptions

1. Each prescription is issued for a specific medication, and is characterized by dosage (unit and amount), frequency, start date, duration (in days), and notes.
2. Dosage unit is one of: mcg, mg, g, ml, IU, tablets, drops.
3. Frequency is one of: once daily, twice daily, three times daily, every other day, weekly, as needed.
4. Only one prescription per medication can be issued within a single appointment.
5. At least one of dosage or notes must be provided (both cannot be empty).
6. Prescriptions for medication can be issued only during an appointment.

1.2.8 Medications

1. Each medication is characterized by a unique name, form, and an optional description.
2. A medication's form is one of: solid, liquid oral, injectable, or topical.
3. Each medication category is characterized by a unique name.
4. Each medication belongs to exactly one medication category.
5. The catalogue of medications must be maintained by the administrator.
6. The directory of medication categories must be maintained by the administrator.

1.2.9 Payments

1. Each payment is characterized by a date and time, and an amount.
2. A payment can be made by cash, card, or bank transfer.
3. A card payment is characterized by a unique transaction number and the last 4 digits of the card number.
4. A bank transfer is characterized by a unique transaction number and a sender bank account number.
5. Each completed appointment must be paid with a single payment.
6. A payment is made only after the completion of the appointment.
7. The total price of an appointment is the sum of prices of all performed procedures and can only be calculated once the appointment is completed.
8. Cash and bank transfer payments must be processed and registered in the system manually by the receptionist.
9. Card payments must be processed through an external payment terminal and registered in the system automatically.

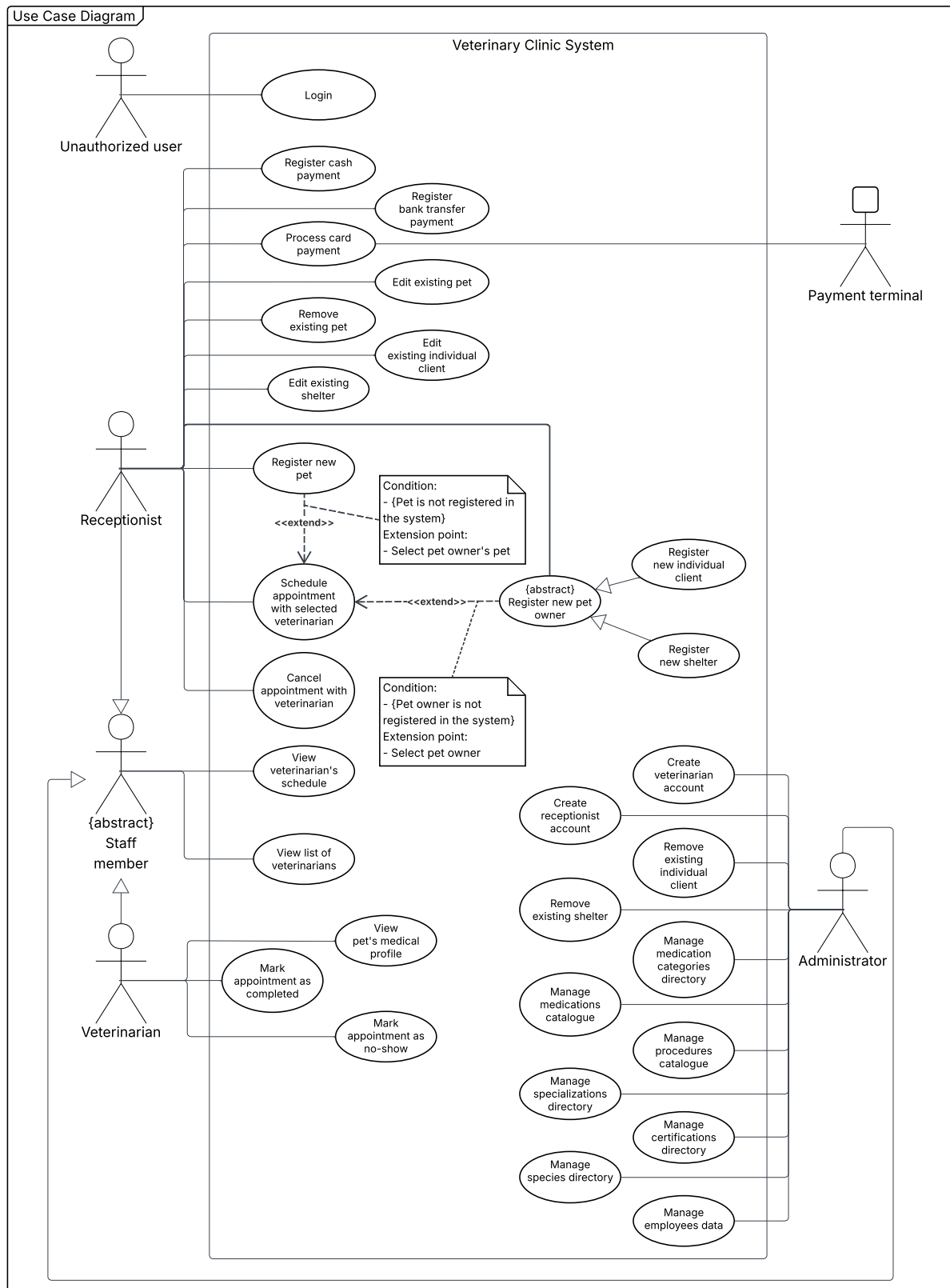
1.2.10 Users

1. Administrator must be able to create accounts for other users who interact with the system.
2. Users interacting with the system (receptionists, veterinarians) must log in to accounts created by the administrator.

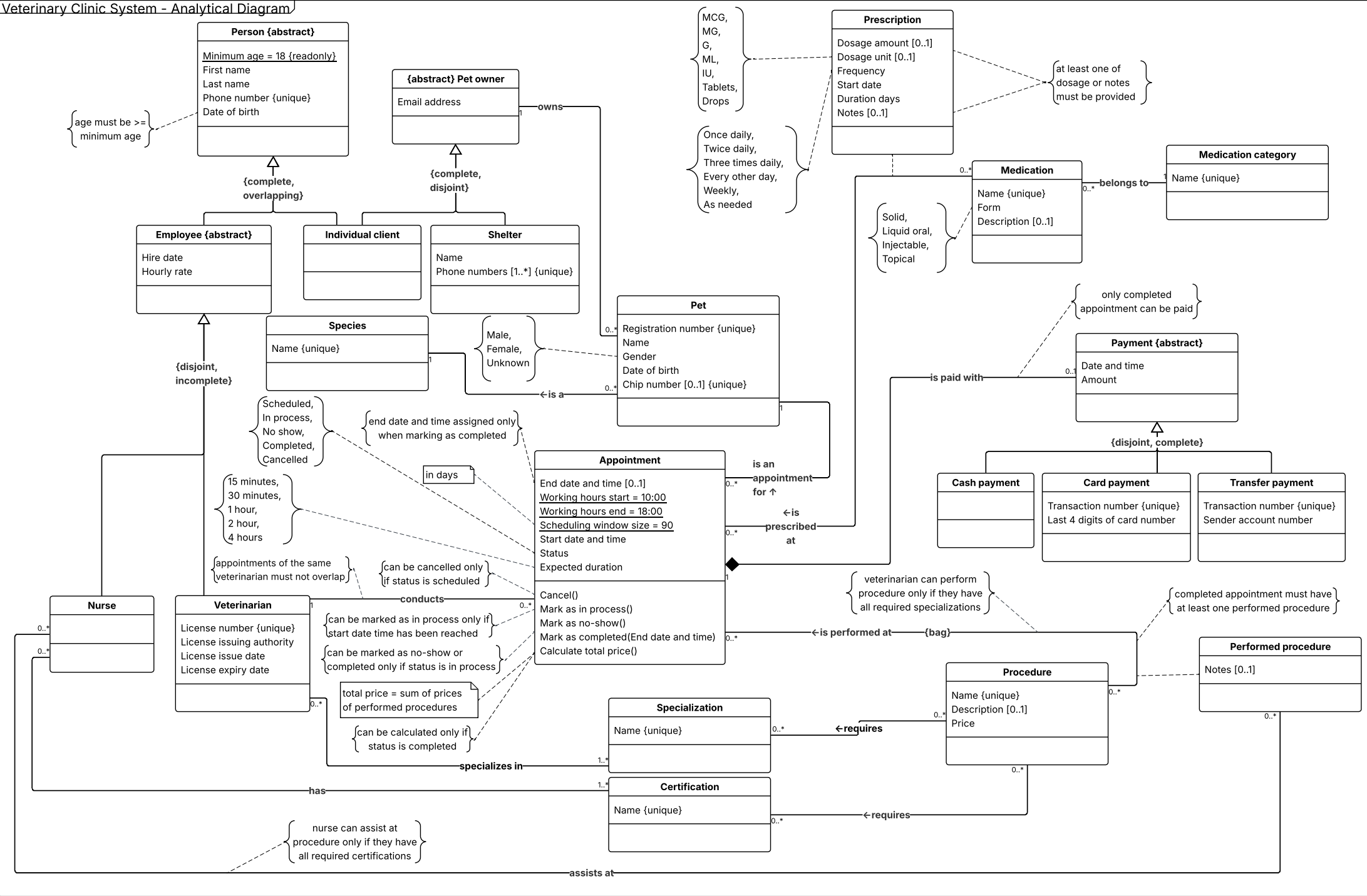
1.3 Non-functional requirements

1. The system shall respond to user requests within 3 seconds for at least 95% of operations under normal workload conditions.
2. The system shall support at least 30 concurrent users without significant degradation of performance.
3. The system shall save appointment scheduling data within 3 seconds after user confirmation.
4. The system shall maintain an availability of at least 99% during clinic working hours, excluding scheduled maintenance.
5. The system shall provide role-based access control to restrict users according to their roles.
6. User passwords shall be stored in a hashed and secure format and shall never be stored as plain text.
7. The system shall ensure referential integrity of data.
8. The system shall perform automatic database backups at least once every 24 hours.
9. The system shall maintain logs of critical user actions, including creation, modification, and deletion of appointments, prescriptions, payments.
10. The system shall provide clear and informative validation and error messages when invalid data is entered or an operation fails.
11. A new employee shall be able to perform basic system operations after no more than 1 hour of training.

2 Use case diagram



Veterinary Clinic System - Analytical Diagram



3 Use cases

3.1 Register new individual client

Register new individual client - Use case scenario

Actor:

Receptionist

Purpose and context:

Receptionist wants to register a new individual client.

Assumptions:

1. The receptionist can cancel the process at any step.

Initiating business event (trigger):

The receptionist initiates individual client registration by clicking "Register new individual client".

Basic flow of events:

1. The receptionist initiates individual client registration by clicking "Register new individual client"
2. The system displays a registration form.
3. The receptionist enters the client's first name, last name, date of birth, phone number, and email address.
4. The receptionist confirms the registration by clicking "Confirm".
5. The system validates the entered data:
 - email format is valid
 - phone number format is valid
 - phone number is not already registered in the system
 - date of birth indicates the client is at least 18 years old
6. The system creates a new individual client.

Alternative flows of events:

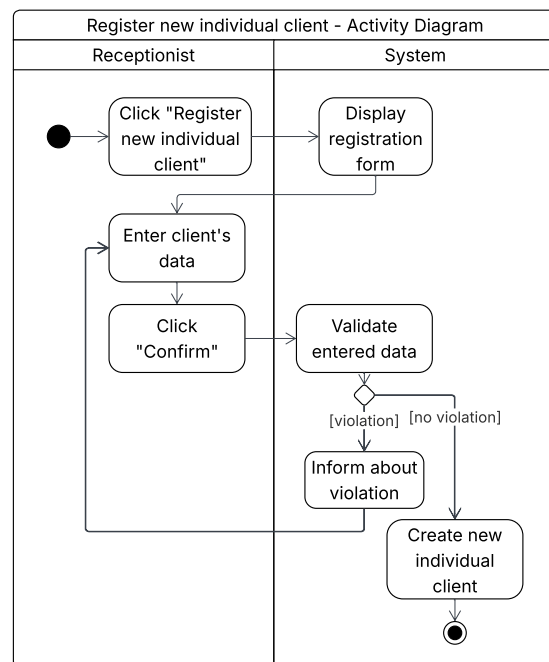
5a. Validation fails

- 5a1. The system informs the receptionist about specific validation error.

- 5a2. Return to step 3.

Post-conditions:

A new individual client is registered in the system.



3.2 Register new shelter

Register new shelter - Use case scenario

Actor:

Receptionist

Purpose and context:

Receptionist wants to register a new shelter.

Assumptions:

1. The receptionist can cancel the process at any step.

Initiating business event (trigger):

The receptionist initiates shelter registration by clicking "Register new shelter".

Basic flow of events:

1. The receptionist initiates shelter registration by clicking "Register new shelter".
2. The system displays a registration form.
3. The receptionist enters the shelter's name, email address, and one or more phone numbers.
4. The receptionist confirms the registration by clicking "Confirm".
5. The system validates the entered data:
 - email format is valid
 - phone numbers format is valid
 - phone numbers are not already registered in the system
6. The system creates a new shelter.

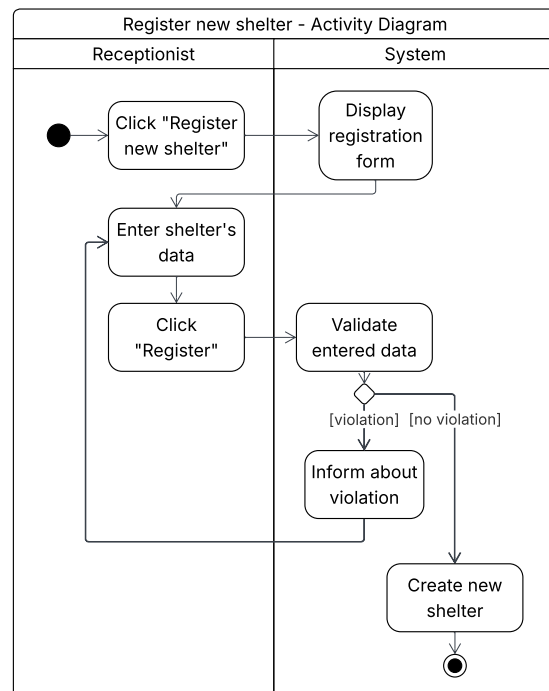
Alternative flows of events:**5a. Validation fails**

5a1. The system informs the receptionist about specific validation error.

5a2. Return to step 3.

Post-conditions:

A new shelter is registered in the system.



3.3 Register new pet

Register new pet - Use case scenario

Actor:

Receptionist

Purpose and context:

Receptionist wants to register a new pet belonging to an existing pet owner (individual client or shelter).

Assumptions:

1. The receptionist can cancel the process at any step.

Pre-conditions:

1. A pet owner (individual client or shelter) exists in the system.
2. A pet owner is selected.

Initiating business event (trigger):

The receptionist initiates pet registration by clicking "Register new pet".

Basic flow of events:

1. The receptionist initiates pet registration by clicking "Register new pet".
2. The system requests pet information.
3. The receptionist enters the pet's name, date of birth, selects the species, gender, and optionally provides a chip number.
4. The receptionist confirms the registration by clicking "Confirm".
5. The system validates the entered data.
6. The system creates a new pet linked to the selected owner.

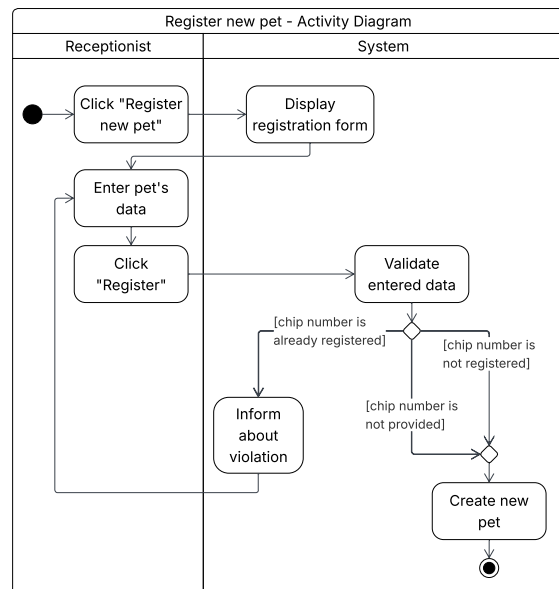
Alternative flows of events:

5a. Chip number is already registered in the system

- 5a1. The system informs the receptionist that the chip number is already registered.
- 5a2. Return to step 3.

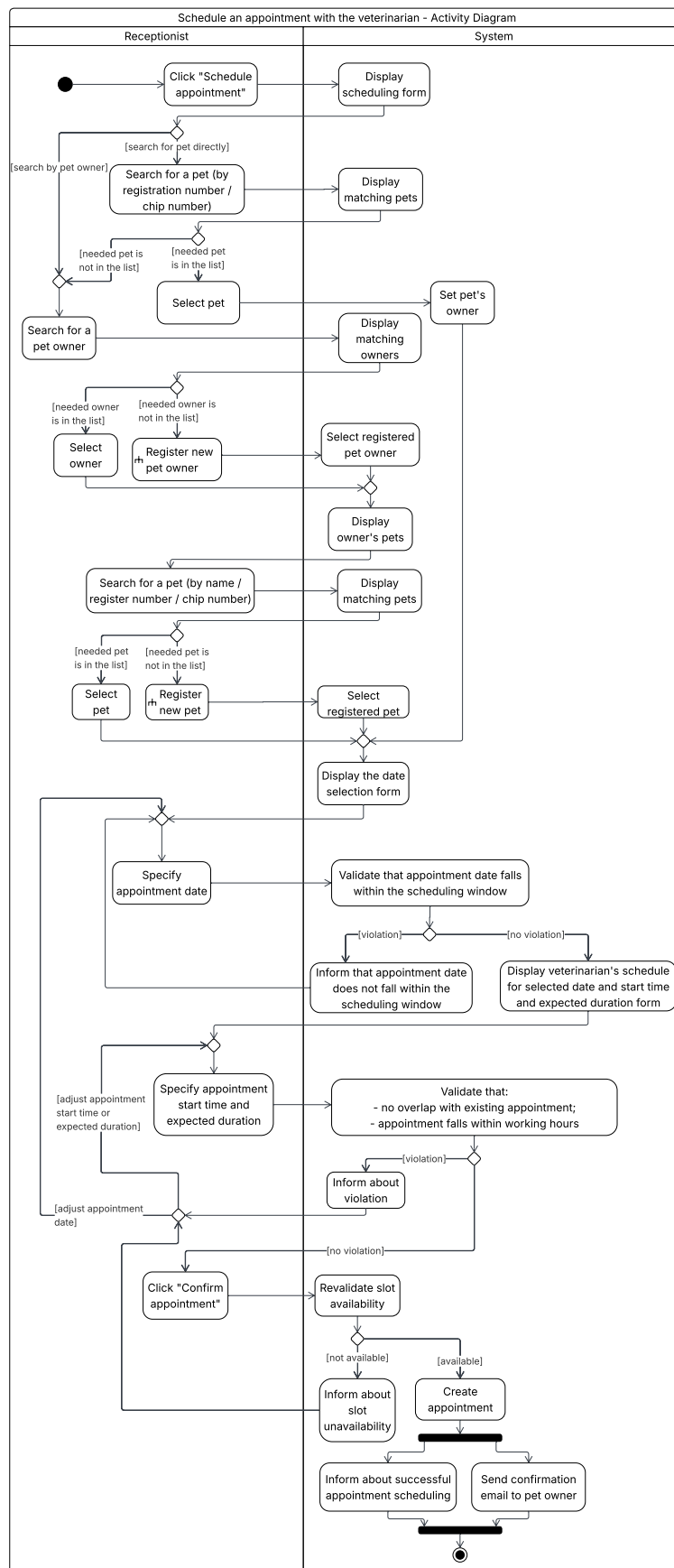
Post-conditions:

A new pet is registered in the system.

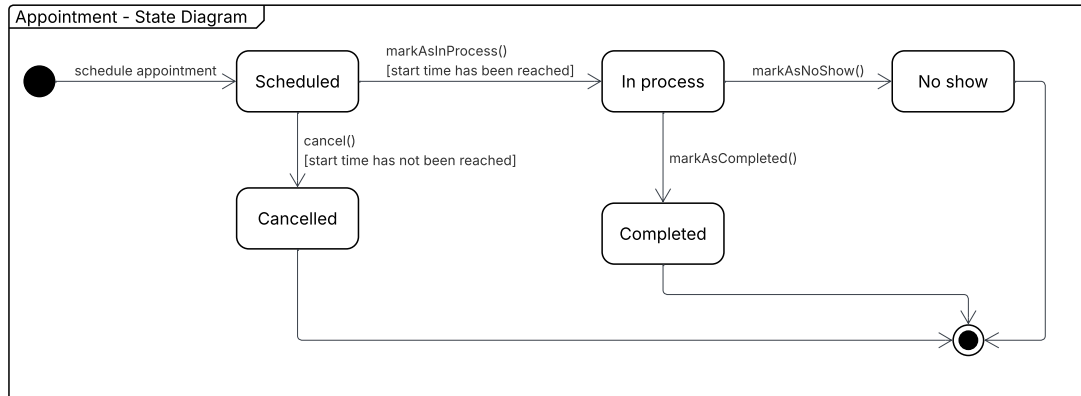


3.4 Schedule appointment with selected veterinarian

Schedule appointment with selected veterinarian - Use case scenario
Actor: Receptionist
Purpose and context: Receptionist wants to schedule an appointment for a pet with the selected veterinarian.
Assumptions: 1. The receptionist can cancel the process at any step. 2. The selected veterinarian has available slots within the scheduling window.
Preconditions: 1. At least one veterinarian exists in the system. 2. A veterinarian is selected.
Initiating business event (trigger): The receptionist initiates appointment scheduling by clicking "Schedule appointment".
Basic flow of events: 1. The receptionist initiates appointment scheduling by clicking "Schedule appointment". 2. The system requests appointment information by displaying a form. 3. The receptionist searches for a pet owner (by name / phone number). 4. The system displays matching owners. 5. The receptionist selects an owner. 6. The system displays the owner's pets. 7. The receptionist searches for a pet (by name / register number / chip number). 8. The system displays matching pets. 9. The receptionist selects a pet. 10. The system displays the appointment date selection form. 11. The receptionist specifies the appointment date. 12. The system validates that the appointment date falls within the scheduling window. 13. The system displays the veterinarian's schedule for selected date, and start time and expected duration form. 14. The receptionist specifies the appointment start time and the expected appointment duration. 15. The system validates that: - the appointment interval does not overlap with another appointment; - the appointment time falls within working hours. 16. The receptionist confirms appointment creation by clicking "Confirm appointment". 17. The system revalidates that the selected appointment interval is still available. 18. The system creates an appointment with status "Scheduled". 19. The system informs the receptionist about successful appointment scheduling and sends a confirmation email to the pet owner.
Alternative flows of events: <i>3a. The receptionist chooses to search for a pet directly</i> 3a1. The receptionist searches for a pet (by register number / chip number). 3a2. The system displays matching pets. 3a3. The receptionist selects a pet. 3a4. The system sets the pet owner automatically. 3a5. Return to step 10. <i>3a3a. Needed pet is not in the list - the receptionist switches to owner search</i> 3a3a1. Return to step 3. <i>5a. Needed owner is not in the list - no matching individual client or shelter is found</i> 5a1. The receptionist registers an individual client or shelter (UC: Register new pet owner). 5a2. The system selects the registered pet owner automatically. 5a3. Return to step 6. <i>9a. Needed pet is not in the list - the individual client or shelter does not have the needed pet</i> 9a1. The receptionist registers a pet (UC: Register new pet) belonging to the individual client or shelter. 9a2. The system selects the registered pet automatically. 9a3. Return to step 10. <i>12a. The appointment date does not fall within the scheduling window</i> 12a1. The system informs the receptionist that appointment date does not fall within the scheduling window. 12a2. Return to step 11. <i>15a. Selected appointment interval overlaps with existing appointment or does not fall within working hours</i> 15a1. The system informs the receptionist that: • the appointment overlaps with another appointment or • the appointment does not fall within working hours. 15a2. Return to the relevant scheduling step (11 for appointment date, 14 for start time and expected duration). <i>17a. Appointment slot becomes unavailable during creation</i> 17a1. The system informs the receptionist that another appointment was scheduled for the selected time. 17a2. Return to the relevant scheduling step (11 for appointment date, 14 for start time and expected duration).
Business rules: B1: Scheduling window size: 90 days. B2: Working hours: 10:00 - 18:00.
Post-conditions: A new appointment with status "Scheduled" is created.



4 Appointment – State diagram



5 Results of dynamic analysis

Dynamic analysis of the use case "Schedule appointment with selected veterinarian" identified the following operations that were not present on the analytical diagram and must be added to the design diagram:

- **Scheduling window validation** (step 12 in "Schedule appointment with selected veterinarian" use case specification) – a static method `Is within scheduling window(Date)` must be added to the `Appointment` class to verify that the selected date does not exceed the scheduling window.
- **Working hours validation** (step 15 in "Schedule appointment with selected veterinarian" use case specification) – a static method `Is within working hours(Start time, Duration)` must be added to the `Appointment` class to verify that the selected appointment interval falls within working hours.
- **Collision check** (step 15 in "Schedule appointment with selected veterinarian" use case specification) – an instance (object) method `Overlaps with (Start date and time, Duration)` must be added to the `Appointment` class to verify that the selected time slot does not overlap with an existing appointment of the same veterinarian.

These methods are reflected in the design diagram.

6 Discussion of design decisions

6.1 Technology choices

The system will be implemented in Java using the Spring Boot framework. Spring Boot provides dependency injection, auto-configuration, and MVC architecture, simplifying integration between the persistence, service, and presentation layers.

Spring Data JPA with Hibernate as underlying JPA provider will be used for persistence. It allows to generate repository implementations from interface definitions, eliminating manual session and transaction management. H2 in file-based mode will serve as the database – it does not require an external server, and persists data between restarts.

Thymeleaf will be used as the server-side template engine for the web GUI. It natively integrates with Spring Boot and does not require a separate frontend build process.

6.2 Implementation of UML constructs

6.2.1 Overlapping inheritance – Person

The overlapping inheritance between **Individual client** and **Employee** is implemented using the flattening approach. Both types are merged into a single **Person** class, which serves as the superclass containing all invariants.

Since a person can simultaneously hold multiple types (e.g., be both an employee and a client), a **PersonRole** enum and **Set<PersonRole>** discriminator was introduced. This allows storing multiple roles at the same time.

Role-specific attributes (**hireDate**, **hourlyRate** for **Employee**, **emailAddress** for **Individual client**) are kept as optional attributes in the same class. Also the following constraints are enforced:

1. If the discriminator contains **Employee**, the attributes **hireDate** and **hourlyRate** are mandatory.
2. If the discriminator contains **Individual client**, the attribute **emailAddress** is mandatory.

6.2.2 Multi-inheritance – Person and Pet owner

Since the overlapping inheritance was flattened into a single **Person** class, there is no longer a need to use an interface to simulate multiple inheritance (which does not exist in Java). Instead, **Person** directly inherits from abstract class **Pet owner**.

However, a new constraint is introduced:

- A **Person** can only own pets if the **Individual client** role is present in their discriminator.

An alternative approach would have been to define **Pet owner** as an interface. However, Hibernate does not support interfaces as mapped entities, which would require two separate associations from **Pet** – one to **Person**, another to **Shelter** – with a {xor} constraint enforcing that exactly one owner is set at a time. To avoid this complexity, **Pet owner** was implemented as an abstract class, allowing a single association from **Pet** to **Pet owner**.

6.2.3 Association classes

Association classes are converted into regular classes. Each instance of such a class holds a reference to exactly one object on each side of the association, preserving the original multiplicity.

However, converting an association class into a regular class introduces a problem: in a UML association, only one link can exist between any two specific objects (unless {bag} or {history} constraints are used). After the conversion, this property is lost – multiple objects of the converted class can be created referencing the same pair, allowing duplicate links.

In the case of **Performed procedure**, the {bag} constraint was applied on the original association, meaning multiple **Performed procedures** for the same **Appointment** and **Procedure** combination are permitted. Therefore, no additional constraint is needed after the conversion.

In the case of **Prescription**, the semantics are different – only one prescription per medication can be issued within a single appointment. Since this property is lost after the conversion to a regular class, a new constraint is added:

- A **Medication** can be prescribed only once per **Appointment**.

6.2.4 Associations

All associations will be implemented using object references in Java. Where the multiplicity starts from 0, references will be set after object creation. Where the multiplicity starts from 1, the referenced objects must be provided as constructor arguments, ensuring the reference is established at the moment of object initialization.

Associations will be implemented as bidirectional, meaning both sides maintain a reference to each other. This is achieved in Java through Hibernate, as relational databases only support unidirectional relationships via foreign keys. Where it makes business sense, a link can be established from both sides – for example, a `Pet` can be assigned to `Pet owner` via `pet.setOwner(owner)` or via `owner.addPet(pet)`.

6.2.5 Enumerations

Java natively supports enumerations, so UML enumerated values are directly converted into Java enums. In the simplest case this is a straightforward mapping – each UML enumerated value becomes a Java enum constant.

One enum in the system goes beyond simple constants: `Appointment duration` carries additional attributes – a `Duration` represents the actual time span and a `Name` will be used for display purposes. Those attributes are defined in the enum constructor and accessed via getters, which Java enums fully support.

6.2.6 Attributes' data types

UML data types are mapped to Java types as follows:

- Strings are mapped to `String`.
- Dates are mapped to `LocalDate`, and date-time values to `LocalDateTime`, and time-only values to `LocalTime`.
- Integer values are mapped to `int`.
- Floating point numbers are mapped to `BigDecimal` rather than `float` or `double`, as `BigDecimal` provides exact decimal arithmetic – this is crucial for monetary values such as prices.
- Duration is mapped to `Duration`.
- Multivalue attributes are mapped to `Set`, since in all cases within the system duplicate values have no business meaning.
- Optional attributes are handled allowing `null` to be passed in constructors and setters.

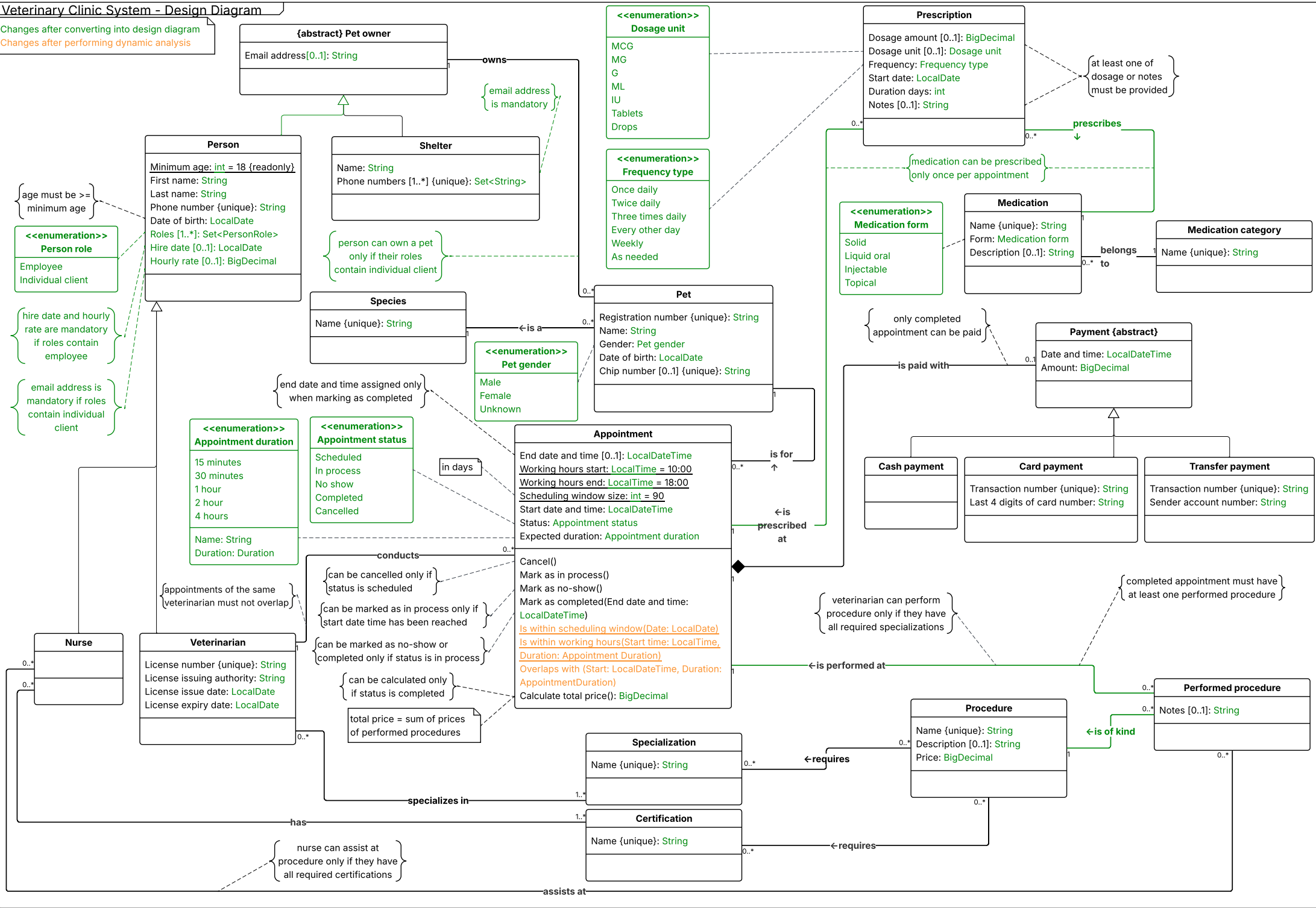
6.2.7 Constraints

- **{unique}** – implemented via `unique = true` parameter in Hibernate `@Column` annotation. Applied to attributes such as phone number, registration number, chip number, license number, names in catalogues.
- **{readonly}** – implemented via `final` keyword in Java.
- **Age >=18** – enforced in the `Date of birth` setter.
- **Hire date and hourly rate are mandatory if roles contain Employee** – enforced in the constructor: when `Employee` role is added, both `hireDate` and `hourlyRate` are required to be set.
- **Email address is mandatory if roles contain Individual client** – enforced in the constructor: when `Individual client` role is added, `email` is required to be set.
- **Person can own a Pet only if roles contain Individual Client** – enforced by checking for the `Individual client` role before adding or removing a link, throwing an exception if the role is absent.
- **Person – Roles [1..*]** – checked in the constructor: at least one role must be present, otherwise an exception is thrown.
- **Shelter – Email address is mandatory, Phone numbers [1..*]** – checked in the constructor.
- **Appointments of the same Veterinarian must not overlap** – checked in the `Appointment service` before `Appointment` creation by iterating through veterinarian's appointments and verifying that the selected time slot does not overlap with any of them.

- **End date and time assigned only when marking as completed** – End date and time is set exclusively inside `Mark as completed()`.
- **Status transition constraints** – enforced inside the dedicated status-changing methods: `Cancel()` checks that the current status is `Scheduled`; `Mark as in process()` checks that the start time has been reached; `Mark as no show()` and `Mark as completed()` check that the current status is `In process`. Additionally, `Mark as completed` checks that there is at least one `Performed` procedure associated with the `Appointment`, and that the provided `End date and time` is after `Start date and time`.
- **Payment can only be set if Appointment status is Completed** – enforced in `Payment` constructor: if the status is not `Completed`, an exception is thrown.
- **At least one of dosage or notes must be provided in Prescription** – checked in `Prescription` constructor: if both dosage and notes are null, an exception is thrown.
- **Veterinarian can perform Procedure only if Veterinarian specializes in all required Specializations** – checked in `Performed procedure` constructor: the `Veterinarian's` specializations are compared against `Procedure's` required specialization, an exception is thrown if any are missing.
- **Nurse can assist at Procedure only if Nurse has all required Certifications** – checked in `Performed procedure` constructor: the `Nurse's` certifications are compared against the `Procedure's` required certifications.
- **Medication can be prescribed only once per Appointment** – checked in `Prescription` constructor by iterating over existing prescriptions of the appointment and verifying that no prescriptions for the same medication already exists.

Veterinary Clinic System - Design Diagram

Changes after converting into design diagram
Changes after performing dynamic analysis



7 GUI Design

7.1 Schedule appointment with selected veterinarian

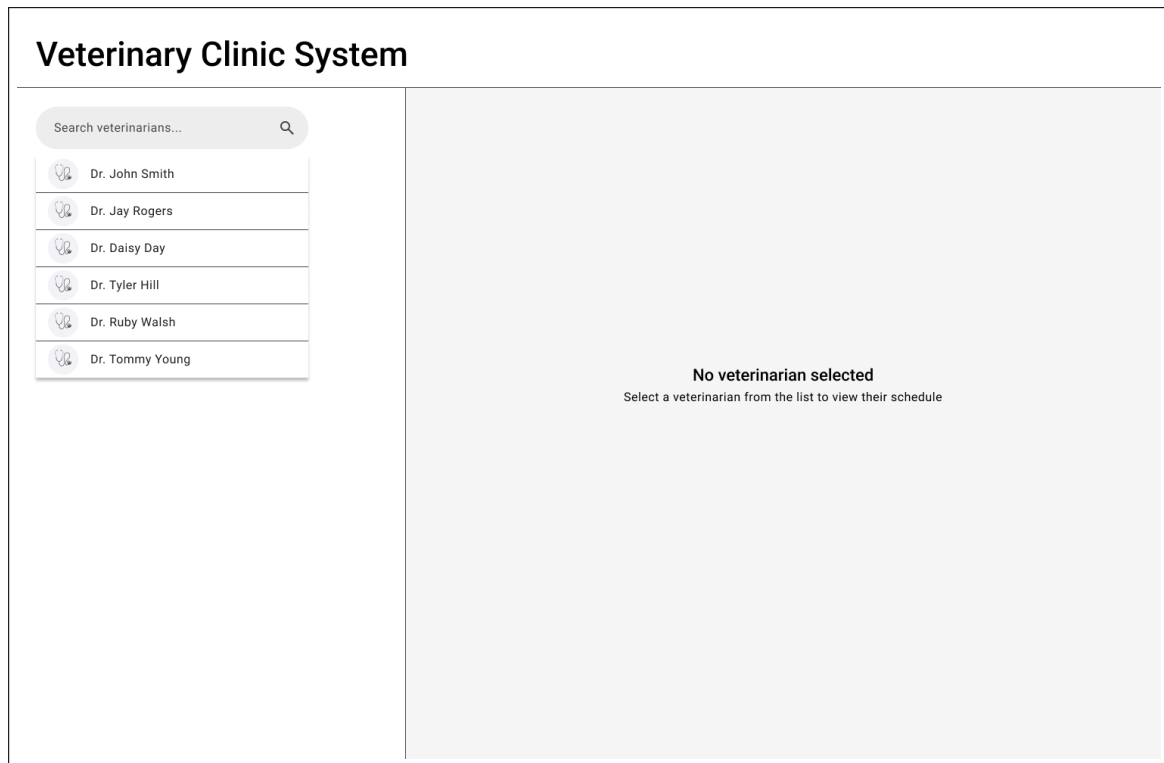


Figure 10: Step 1. Main page

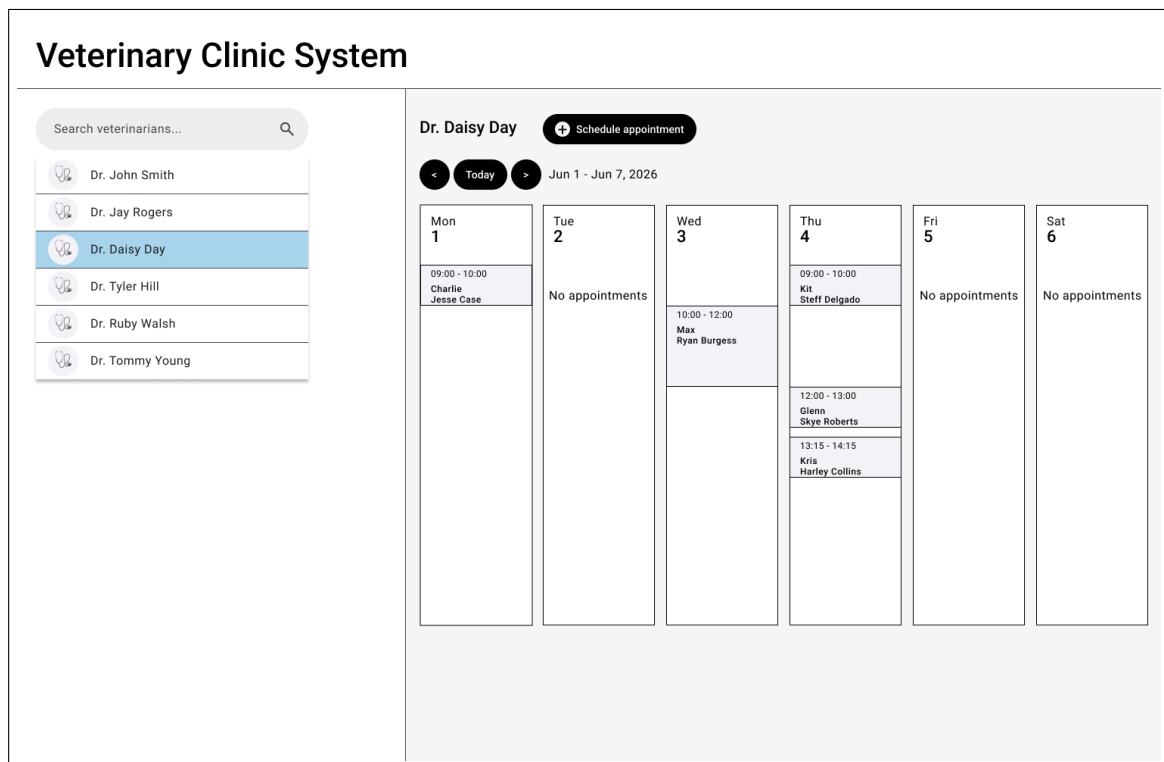


Figure 11: Step 2. Main page - veterinarian selected

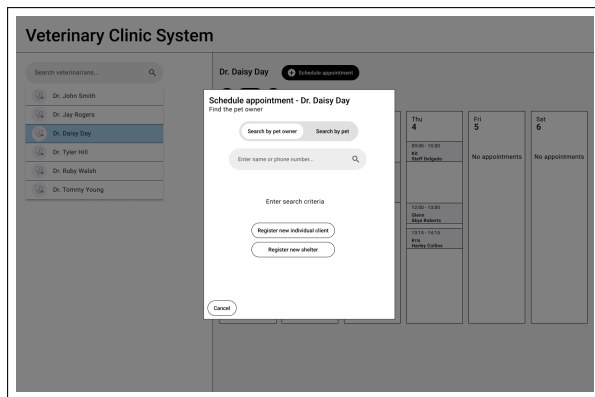


Figure 12: Step 3. Search by pet owner (main flow)

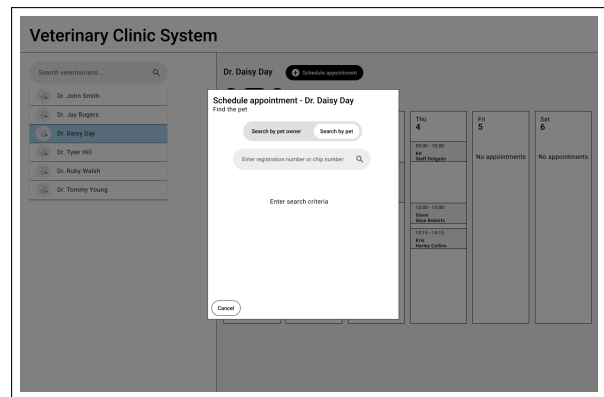


Figure 13: Step 3. Search for pet directly (alternative flow)

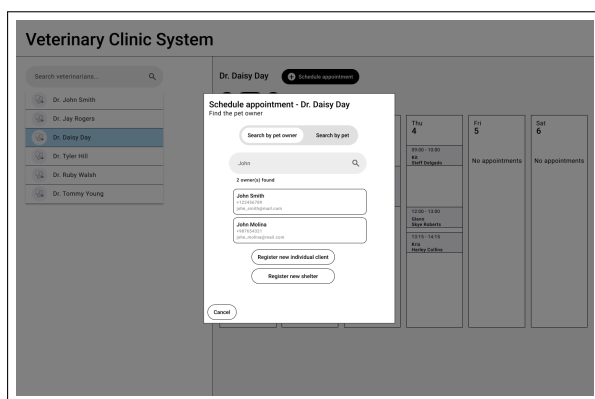


Figure 14: Step 4. Found pet owners (main flow)

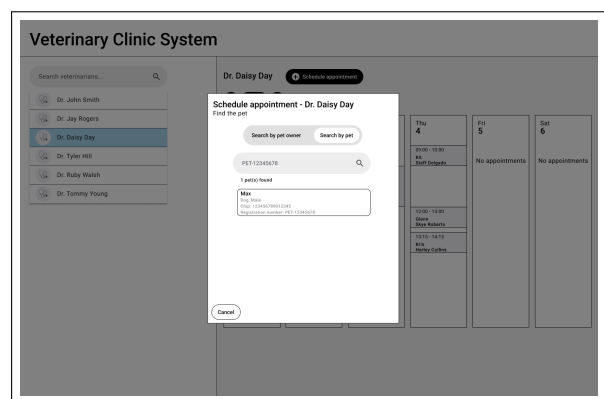


Figure 15: Step 4. Found pets (alternative flow)

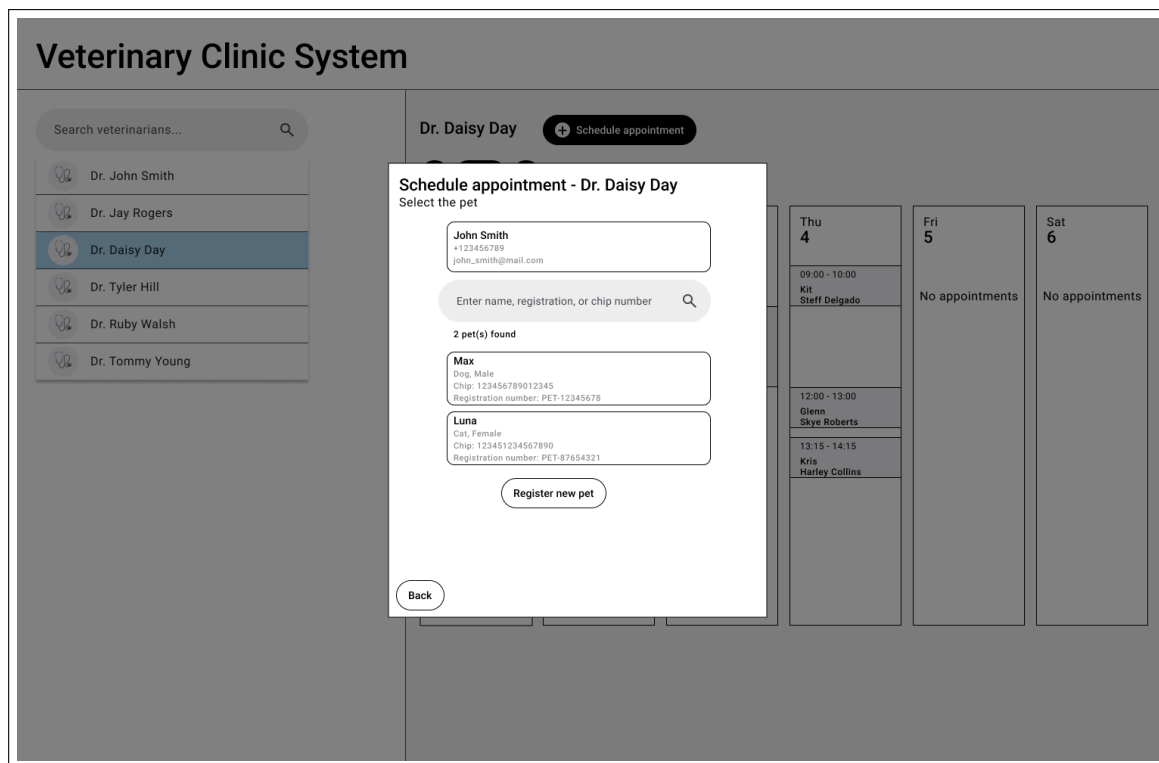


Figure 16: Step 5. Owner chosen - owner's pets displayed

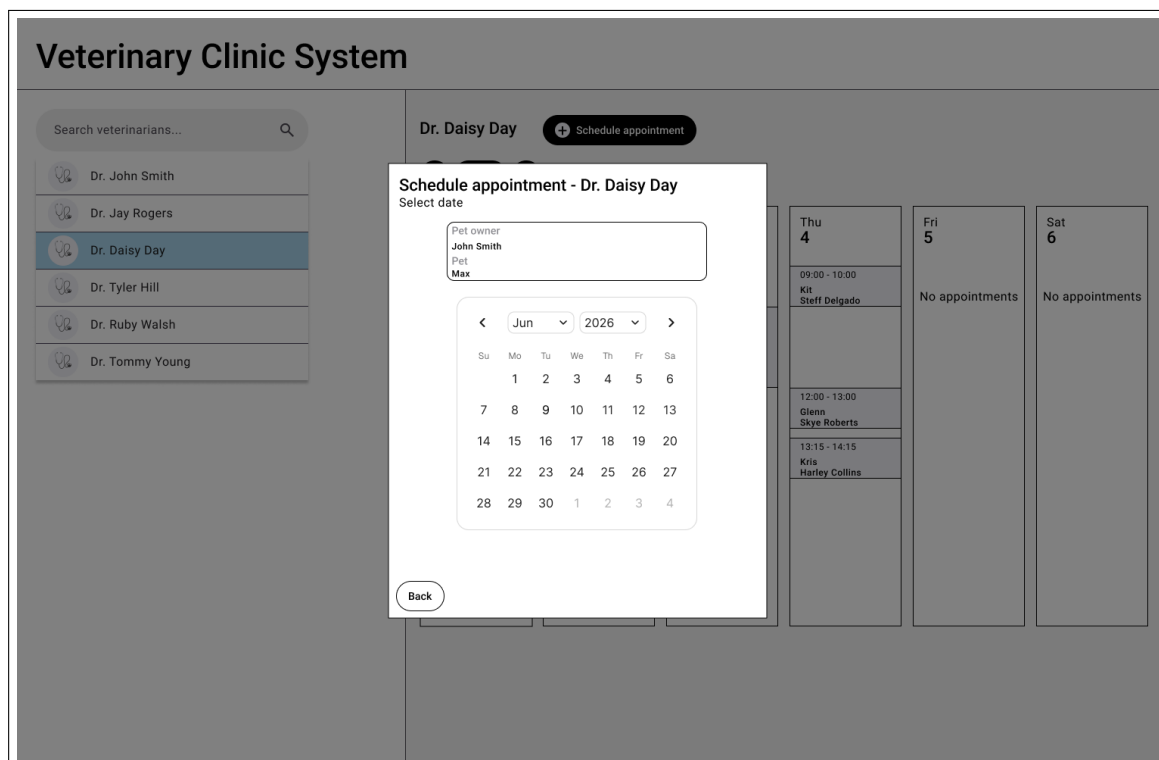


Figure 17: Step 6. Pet chosen - calendar displayed

Veterinary Clinic System

Search veterinarians...

Dr. John Smith

Dr. Jay Rogers

Dr. Daisy Day

Dr. Tyler Hill

Dr. Ruby Walsh

Dr. Tommy Young

Schedule appointment - Dr. Daisy Day

Select time and duration

Pet owner
John Smith
Pet
Max

Jun
2026

Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4

Existing appointments
10:00 - 10:30 (30 min)
12:00 - 13:00 (1 hour)

Start time
01:00 PM

Expected duration
30 minutes

☒ Time slot available
13:00 - 13:30 (30 minutes)

Back Confirm

Thu 4
09:00 - 10:00
Kit
Steff Delgado
12:00 - 13:00
Glenn
Skye Roberts
13:15 - 14:15
Kris
Harley Collins

Fri 5
No appointments

Sat 6
No appointments

Figure 18: Step 7. Date chosen - appointment details requested

Veterinary Clinic System

Search veterinarians...

Dr. John Smith

Dr. Jay Rogers

Dr. Daisy Day

Dr. Tyler Hill

Dr. Ruby Walsh

Dr. Tommy Young

Dr. Daisy Day
Schedule appointment

Today
Jun 1 - Jun 7, 2026

Mon 1 09:00 - 10:00 Charlie Jesse Case	Tue 2 No appointments	Wed 3 10:00 - 12:00 Max Ryan Burgess	Thu 4 09:00 - 10:00 Kit Steff Delgado 12:00 - 13:00 Glenn Skye Roberts 13:15 - 14:15 Kris Harley Collins	Fri 5 No appointments	Sat 6 No appointments
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Appointment scheduled successfully
Confirmation email sent to john_smith@gmail.com

Figure 19: Step 8. Appointment scheduling confirmed

7.2 Register new individual client

The screenshot shows the 'Veterinary Clinic System' interface. On the left, a sidebar lists veterinarians: Dr. John Smith, Dr. Jay Rogers, Dr. Daisy Day (selected), Dr. Tyler Hill, Dr. Ruby Walsh, and Dr. Tommy Young. The main area displays 'Dr. Daisy Day' with a '+ Schedule appointment' button. A modal form titled 'Schedule appointment - Dr. Daisy Day' is open, with the subtitle 'Find the pet owner'. Inside the modal, the section 'Register new individual client' is active, with the instruction 'Add a new individual client to the system'. The form contains the following fields: 'First name *' and 'Last name *' (text inputs), 'Date of birth *' (with dropdowns for DD, MM, and YYYY), 'Phone number *' (text input), and 'Email address *' (text input). At the bottom of the modal are 'Cancel' and 'Confirm' buttons. In the background, a calendar view shows appointments for Thursday 4th, Friday 5th, and Saturday 6th.

Figure 20: Register new individual client – if client does not exist, can be registered at step 3 by clicking “Register new individual client”

7.3 Register new shelter

The screenshot shows the 'Veterinary Clinic System' interface, similar to Figure 20. The 'Register new shelter' modal form is open, with the instruction 'Add a new shelter to the system'. The form contains the following fields: 'Shelter name *' (text input), 'Email address *' (text input), and 'Phone numbers * (at least one required)' (text input). Below the phone numbers field is a '+ Add phone number' button. At the bottom of the modal are 'Cancel' and 'Confirm' buttons. The background calendar view is also visible, showing appointments for Thursday 4th, Friday 5th, and Saturday 6th.

Figure 21: Register new shelter – if shelter does not exist, can be registered at step 3 by clicking “Register new shelter”

7.4 Register new pet

The screenshot displays the 'Veterinary Clinic System' interface. On the left, a sidebar lists veterinarians: Dr. John Smith, Dr. Jay Rogers, Dr. Daisy Day (selected), Dr. Tyler Hill, Dr. Ruby Walsh, and Dr. Tommy Young. The main area shows 'Dr. Daisy Day' with a 'Schedule appointment' button. A modal form titled 'Register new pet' is open, prompting the user to 'Add a new pet to the system'. The form includes fields for 'Pet name *', 'Species *' (with a dropdown), 'Gender *' (with a dropdown), 'Date of birth *' (with DD, MM, and YYYY input boxes), and 'Chip number *'. At the bottom of the modal are 'Cancel' and 'Confirm' buttons. In the background, a calendar view shows appointments for Thursday, Friday, and Saturday.

Thu 4	Fri 5	Sat 6
09:00 - 10:00 Kit Steff Delgado	No appointments	No appointments
12:00 - 13:00 Glenn Skye Roberts		
13:15 - 14:15 Kris Harley Collins		

Figure 22: Register new pet – if pet does not exist, can be registered at step 5 by clicking “Register new pet”